ECE 310 Fall 2026
Lecture 9: Transfer functions and LTI system response: Part 1

Corey Snyder

University of Illinois Urbana-Champaign

September 14th, 2026



Learning Objectives

- Compute the response of an LTI system to an input signal using the z-transform.
- Write the transfer function of an LTI system described by an LCCDE.
- Identify FIR and IIR systems based on characteristics of an LCCDE and its transfer function.



LTI System Response and the z-transform

For an LTI system with impulse response $h[n]$, let $H(z)$ denote the system's transfer function. We know thus far that

$$y[n] = x[n] * h[n].$$

Now, we may express the input-output relationship of LTI systems a second way using the convolution property of the z-transform:

$$Y(z) = X(z)H(z), \ \text{ROC} \ = \ \text{at least} \ R_x \cap R_h.$$

An important consequence of the convolution property is that we see another way to express the transfer function of an LTI system:

$$H(z) = \frac{Y(z)}{X(z)}.$$



Transfer Functions and LCCDEs

Recall from previous lectures how we defined general Linear Constant Coefficient Difference Equations (LCCDE):

$$y[n] + \sum_{k=1}^{N} a_k y[n-k] = \sum_{k=0}^{M-1} b_k x[n-k].$$

What happens if we take the z-transform of both sides?

$$Y(z) + \sum_{k=1}^{N} a_k z^{-k} Y(z) = \sum_{k=0}^{M-1} b_k z^{-k} X(z)$$

$$Y(z)\left(1 + \sum_{k=1}^{N} a_k z^{-k}\right) = X(z)\left(b_k z^{-k}\right)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M-1} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}}$$

(Form #1 from Lecture 8)

Exercise 1: Finding Impulse Response

Compute the impulse response of the following causal LCCDE:

$a^n u[n] \xleftrightarrow{\mathcal{Z}} \frac{1}{1-az^{-1}}, |z|>|a|$

$$y[n] = \frac{1}{3}y[n-1] + x[n]$$

$\downarrow \mathcal{Z}$

$$Y(z) = \frac{1}{3}z^{-1}Y(z) + X(z)$$

$$Y(z)\left(1 - \frac{1}{3}z^{-1}\right) = X(z)$$

$$\frac{Y(z)}{X(z)} = H(z) = \frac{1}{1-\frac{1}{3}z^{-1}}, \quad |z| > \frac{1}{3}$$

$\downarrow \mathcal{Z}^{-1}$

$$h[n] = \left(\frac{1}{3}\right)^n u[n]$$

h[n] is right-sided

Exercise 2: Finding Impulse Response

Compute the impulse response of the following causal LCCDE: menti.com: 2105 8661

$a^n u[n] \xleftrightarrow{\mathcal{Z}} \frac{1}{1-az^{-1}}, |z|>|a|$

$$y[n] = \frac{1}{3}y[n-1] + x[n] - 3x[n-1] + x[n-2]$$

$\downarrow \mathcal{Z}$

$$Y(z) = \frac{1}{3}z^{-1}Y(z) + X(z) - 3z^{-1}X(z) + z^{-2}X(z)$$

$$Y(z)\left(1-\frac{1}{3}z^{-1}\right) = X(z)\left(1-3z^{-1}+z^{-2}\right)$$

$$\frac{Y(z)}{X(z)} = H(z) = \frac{1-3z^{-1}+z^{-2}}{1-\frac{1}{3}z^{-1}}, \quad |z|>\frac{1}{3}$$

$$= \frac{1}{1-\frac{1}{3}z^{-1}} - \frac{3z^{-1}}{1-\frac{1}{3}z^{-1}} + \frac{z^{-2}}{1-\frac{1}{3}z^{-1}}, \quad |z|>\frac{1}{3}$$

$\downarrow \mathcal{Z}^{-1}$

$$h[n] = \left(\frac{1}{3}\right)^n u[n] - 3\left(\frac{1}{3}\right)^{n-1} u[n-1] + \left(\frac{1}{3}\right)^{n-2} u[n-2]$$

Exercise 3: Finding Impulse Response

Compute the impulse response of the following causal LCCDE:

$$y[n] = \frac{1}{2}y[n-1] + \frac{3}{16}y[n-2] + x[n] - 3x[n-1]$$

$$Y(z) = \frac{1}{2}z^{-1}Y(z) + \frac{3}{16}z^{-2}Y(z) + X(z) - 3z^{-1}X(z)$$

$$\frac{Y(z)}{X(z)} = H(z) = \frac{1-3z^{-1}}{1-\frac{1}{2}z^{-1}-\frac{3}{16}z^{-2}} = \frac{1-3z^{-1}}{\left(1+\frac{1}{4}z^{-1}\right)\left(1-\frac{3}{4}z^{-1}\right)}$$

$$\xleftarrow{\mathcal{Z}^{-1}} \frac{\frac{13}{4}}{1+\frac{1}{4}z^{-1}} + \frac{\frac{-9}{4}}{1-\frac{3}{4}z^{-1}} = \text{PFD}$$

$$h[n] = \frac{13}{4}\left(-\frac{1}{4}\right)^n u[n] - \frac{9}{4}\left(\frac{3}{4}\right)^n u[n]$$

Characterizing Transfer Functions

